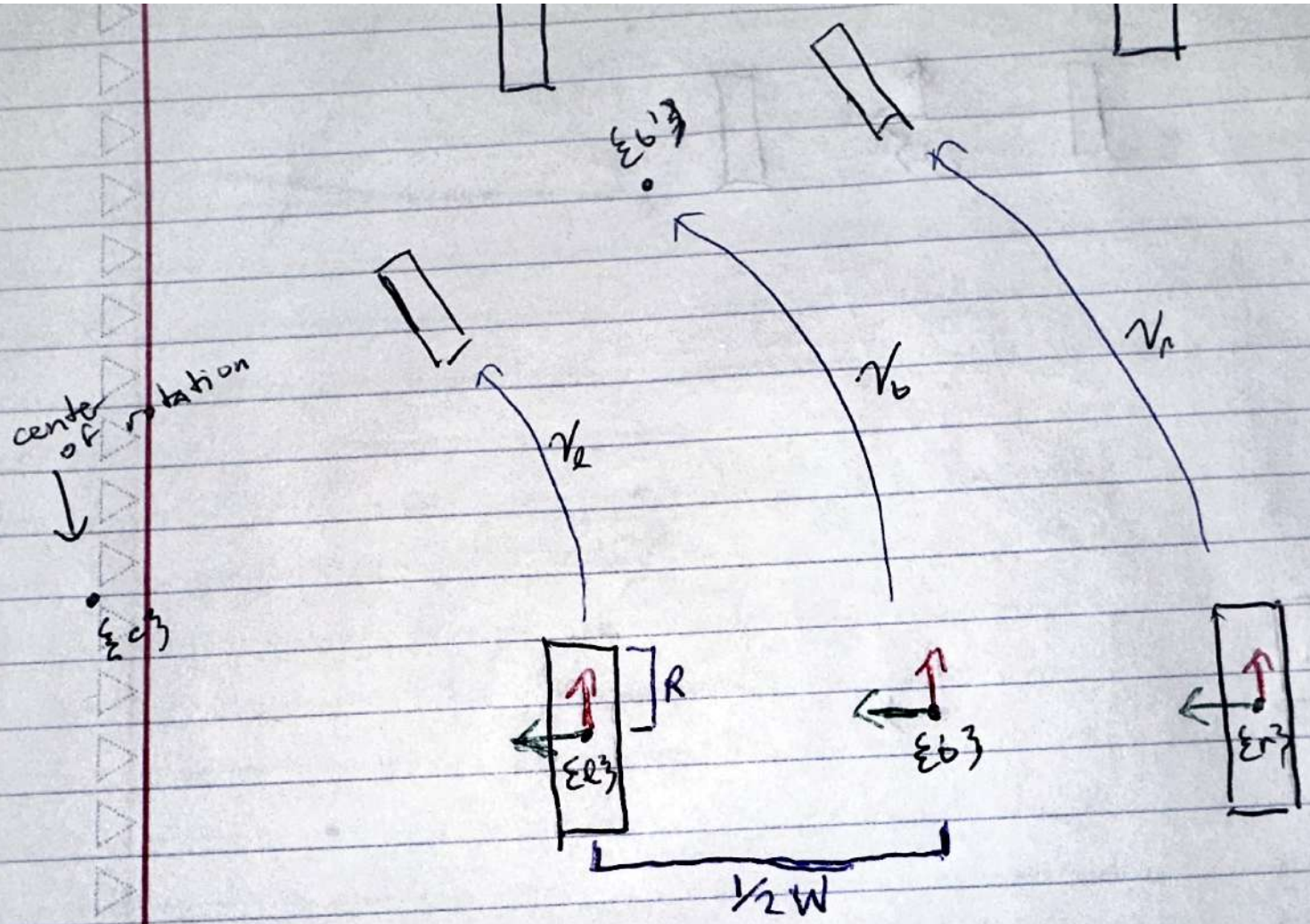
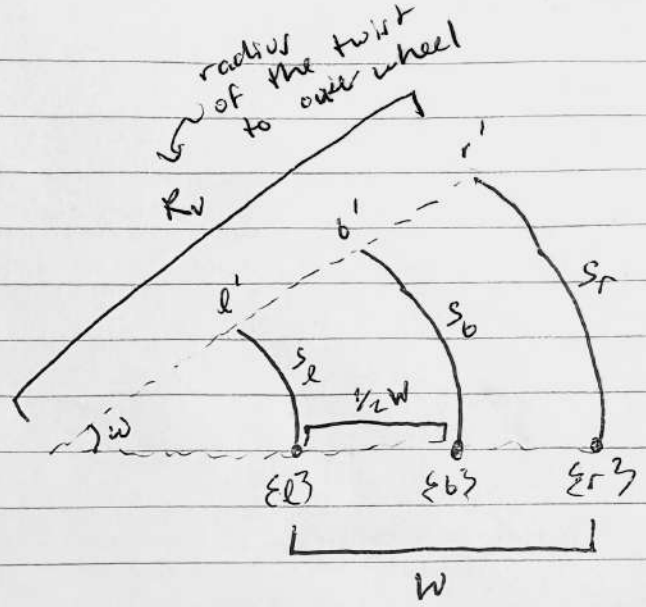

center of rotation
{c}
{b'}
γ_l
γ_b
γ_r
R
{l}
{b}
{r}
1/2 W



S' is arc length

(1A) $\frac{S_r}{S_l} = \frac{R_v}{R_v - W}$ $\qquad \omega = \frac{S_r}{R_v}$

$$S_r(R_v - W) = R_v S_l$$
$$S_r R_v - S_r W = R_v(S_l)$$
$$S_r R_v - S_l R_v = S_r W$$
$$R_v(S_r - S_l) = S_r W$$
$$R_v = \frac{S_r W}{S_r - S_l}$$

$$\omega = \cancel{S_r}\left(\frac{S_r - S_l}{\cancel{S_r} W}\right)$$

$$\boxed{\omega = \frac{S_r - S_l}{W}} \quad Eq(1)$$

due to equidistance between wheels and $\{b\}$ $\qquad S_b = \frac{S_l + S_r}{2} \quad Eq(2)$

Due to diff drive, S_b above $= V_x$ (forward velocity) in body frame

for IK, we know V_x, $V_y = 0$, ω.

~~[illegible] (1A) above~~: Furthermore, for unit twist,